



PraK V4 at the Video Browser Showdown 2026

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Abstract. PraK V4 is an interactive video search system built for retrieval competitions. It introduces five advances over the prior release: (1) spatial conjunction of localized queries, letting users specify AND constraints across separate regions of interest, evaluated in a user study; (2) semi-automated user-interface adaptation for AVS tasks using online learning; (3) single-video browsing through a redesigned player and within-video querying; and (4) a parallelized backend for lower latency, (5) an updated keyframe layout strategy.

Keywords: Interactive Video Retrieval · CLIP · Localized Queries retrieval

1 Introduction

Fast, accurate video search is vital in professional and everyday contexts [18, 29, 30]. To benchmark progress under realistic conditions, the Video Browser Showdown (VBS) annually simulates a broad range of video retrieval tasks and data [6, 15, 29]. As in the previous edition, VBS 2026 will require searching across V3C [22] (everyday scenes), the Marine Video Kit (MVK, underwater footage) [28], and a laparoscopic gynecology (LHE) dataset [17]. Given the scale of these collections and their cross-dataset variability, participating systems typically rely on multimodal foundation models, such as CLIP [19, 23], and perform text-based search over pre-extracted keyframes [29]. This approach works well for everyday life scenes, whose diverse scenes respond to text prompts; however, the highly homogeneous MVK and LHE collections remain challenging for text-only querying. Consequently, search on these datasets often involves extended browsing, with participants manually scanning the pool, raising cognitive load and increasing the risk of misses. Moreover, keyframe-centric methods query single moments only, making it difficult to express complex temporal events. Below, we outline how systems in the previous edition addressed these challenges.

While all winning systems of last years [1, 5, 24] rely on multimodal text queries, NII-UIT [5] and Vibro [24] complementarily utilize object detectors for efficient filtering and querying of keyframes. NII-UIT further employs a large

language model for advanced query expansion. Furthermore, systems also allow for direct image-image queries, either by directly using a keyframe from the database, or by generating images with stable diffusion from textual descriptions [5]. PraKV3 [27], the runner-up of 2024, further includes Bayesian Updates, which rerank a retrieved candidate set on-the-fly with the help of positive and negative examples, and consequently change the order in which keyframes are displayed on the screen. While almost all winning teams support the layout of candidates on a 2D grid in their row-major order, NII-UIT [5], VISIONE [1, 2], and PraK [27] also provide a video-grouped layout, where keyframes are grouped into blocks of videos. In contrast, VIBRO [24] employs a grid layout algorithm [3] to visualize keyframes sorted by color. Most teams include capabilities to browse the whole video by selecting a specified keyframe, for example, with a video player [27].

Although PraK was the runner-up in VBS 2025, we identified some drawbacks in our systems through log analysis and expert interviews. PraK historically underperforms in Ad-Hoc video search, and participants spent vast amounts of time searching for concrete information in an already correctly identified video. Furthermore, while PraK V3 also included localized text queries for MVK and LHE, users missed combining multiple locations. In the following, we will give an overview of the general PraK system, and subsequently provide details about improved features of this year’s iteration.

2 PraK System Architecture and Description

PraK consists of four parts that build on experience and further investigate ideas from VIRET [12], SOMHunter [11, 13], and CVHunter [14]. The four parts comprise: **1)** A preprocessing pipeline that reduces videos to keyframe features. The pipeline first detects shots with TransNet v2 [26] and extracts keyframes via agglomerative clustering. For retrieval, the keyframes are accompanied with corresponding CLIP features finetuned for image–image search [23], plus the same type of features for a 5-region centerpiece partitioning for MVK/LHE localized queries. **2)** A stateless in-memory data service that answers requests (e.g., text queries). **3)** A stateful backend that manages session state (e.g., current candidates) and delegates to the data service as needed. **4)** A web frontend showing keyframes in a 4-column ranked layout, with optional grouping by video (max one keyframe per video). Hovering previews the source video or fetches all its keyframes. The control panel supports text, temporal, and localized text queries (via suffixes like *br* for bottom-right). Bayesian Updates reorder the current result set from clicked positives, so similar keyframes move upward.

3 Added Features for VBS 2026

Section 2 described the PraK system as used in VBS 2025. We will now describe and motivate planned changes for VBS 2026. We do not plan major changes to the UI, but will include options for localized queries (Sect. 3.1 and an option for switching between keyframe layouts (Sect. 3.5), as displayed in Fig. 1.

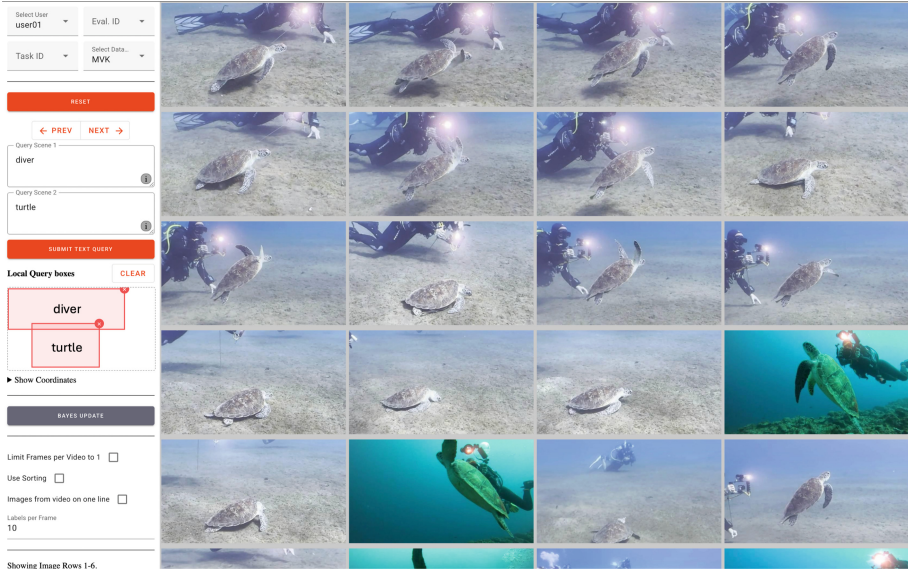


Fig. 1. Screenshot of the UI of PraK V4.

3.1 Spatial Conjunction of Localized Queries

PraK V3 supports sub-region search via a query suffix using a static 5-region grid (four corners + center), following SOMHunter [13]. Jäckl et al. [7, 8] compared this static approach to a dynamic one (object-detected regions) on MVK: users draw a box plus text, and adding a geometric distance (box match) yielded the best effectiveness. Their study, however, was limited to keyframes (image retrieval). We also note that sub-region search and conjunctive queries in SOMHunter were tested on heterogeneous datasets, where a standard whole-image text query can be highly effective.

As a follow up, we extend the studies on homogeneous data [7, 8] to *shots* and allow also *conjunctive* sub-region queries in PraK (multiple boxes per shot matched simultaneously). Five experienced annotators labeled random MVK shots (avg. 12s; range 8–16s), skipping shots where sub-regions were irrelevant. Each shot has one whole-image query and at least one localized query (box+text). In total, we collected 212 annotations and compared five sub-region variants by matching CLIP embeddings between visual regions and text, comparing the following approaches:

- **Whole-Image** – The baseline whole-image text query.
- **Static** – Matches bounding box to best static region based on IoU, sum distances over multiple annotation boxes
- **Dynamic** – Similar to **Static**, but dynamic regions are used.
- **Dynamic with CD** – Distances are computed between all extracted sub-regions and the respective localized query text. At the same time, the

Table 1. Retrieval performance averaged over all (conjunctive) annotations. The recall is presented as percentage (%). We highlight best-performing entries.

	R@1	R@10	R@100	R@1000
Whole-Image	0.04	0.19	0.46	0.73
Static	0.07	0.28	0.59	0.83
Dynamic	0.04	0.21	0.43	0.74
Dynamic with CD	0.13	0.31	0.56	0.81
Dynamic with IoU	0.11	0.29	0.59	0.83

Euclidean L2 distance is computed between the centroid of the annotated bounding box and the centroid of the bounding boxes extracted by object detectors. Both distances are standardized with pre-computed averaged means and standard deviations, and summed up. In a second step, this combined distance is stratified to a per-keyframe distance.

- **Dynamic with IoU** – Same as **Dynamic with CD**, but instead of the centroid distance, IoU is used.

For object detection, Grounded-SAM is used with the same parameters as used by Jäckl et al. [7, 20].

Averaged recall improves substantially over **Whole-Image** for **Static** and both dynamic fusion variants, while **Dynamic** alone does not improve (see Table 1), contrary to Jäckl et al. [7]. First, our bounding boxes are not as accurate as those collected by Jäckl et al. [7], as our annotators were not allowed to draw directly into keyframes, but rather on a separate box based on observed videos. Second, we did not specify that our annotators should query for *objects*, but for interesting regions - consequently, many queries included region descriptions such as *school of fish* that were not detected as one coherent region, but as multiple objects. Although the averaged recall is comparable, we detected that the performance of the approaches strongly depends on the annotation (see Fig. 2). We thus include both options.

We furthermore measure the impact of conjuncting multiple queries. We show results for **Static** in Table 2. While the overall recall is comparable when using all available boxes (*Full* variant), we see a substantial performance decay when randomly choosing 1 box for each annotation (*Random* variant, averaged over 5 runs) - we conclude that some videos strongly profit from the option that one can conjunct multiple boxes as they inherently need multiple annotations (e.g. because there are two spatially distinct objects). However, our experiments are limited as they randomly choose a single box, and the annotator might have written completely different queries if only one box was allowed.

3.2 Intra-video Querying

In previous iterations, PraK provided users with two primary options for searching information within an identified video: a basic video player with a time bar for

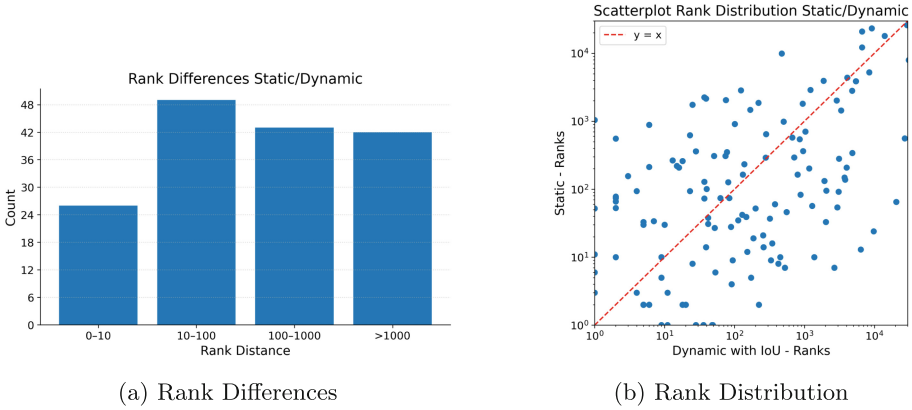


Fig. 2. a) We compute for each annotation the rank differences between the **Static** and **Dynamic with IoU** method. b) We show the exact distribution.

Table 2. Recall-at- k ($R@k$) of **Static** using all boxes (Variant=*Full*) or a single box randomly selected from the available annotations (Variant=*Random*).

#Annot.	Variant	R@1	R@10	R@100	R@1000
1 box	Full/Random	0.06	0.25	0.68	0.87
2 boxes	Full	0.1	0.32	0.62	0.86
	Random	0.05 ± 0.02	0.20 ± 0.03	0.44 ± 0.03	0.72 ± 0.03
3 boxes	Full	0.05	0.26	0.32	0.68
	Random	0.01 ± 0.02	0.04 ± 0.04	0.11 ± 0.06	0.55 ± 0.03

manual navigation, and to display all representative keyframes associated with a video. In the new iteration, we add two popular extensions of this functionality. First, the video player will be enhanced to allow user-defined replay speed. The tool will also incorporate explicit information on scene boundaries, allowing users to skip efficiently between scenes and focus directly on relevant video sections. We emphasize that extending the navigation bar with visual or semantic content is an interesting future work [4, 25]. Second, we extend the querying capabilities to allow users to retrieve information within a single video, for example, by ranking the keyframes of that specific video like in CVHunter [14].

3.3 Parallelized Backend

A formerly single synchronous backend created bottlenecks under concurrent load. V4 deploys multiple asynchronous backend instances behind NGINX with replicated stateless data services and least-connections load balancing. This enables end-to-end parallel processing; in internal tests we handle 25 users in parallel with < 3 s per query.

3.4 Online Learning

In addition to user-driven Bayesian updates, we plan to integrate also a lightweight online automated re-ranking strategy for AVS tasks. PraK can combine the Bayesian model either with the classic Rocchio algorithm [21] efficiently used in dense retrieval systems [10], or Support Vector Machines (SVMs) incrementally trained with batches of submissions of relevant frames [16].

3.5 Keyframe Layouts

A pure 4-column ranked grid is fast for strong candidates but inefficient for ruling out absent targets in homogeneous sets like MVK. We add a hybrid layout motivated by Jäckl et al. [9]: the top-16 keyframes remain rank-ordered in 4 columns; the remaining items are spatially sorted with FLAS [3] using CLIP features for faster exclusion during broad scans. Users can still group by video or show one keyframe per video.

3.6 Additional Changes

Bayesian updates now only affect the on-screen ordering, not underlying scores, ensuring that each update visibly prioritizes the selected positives without accumulating hidden history. We also dropped patch-based texture queries in favor of localized text queries, which proved more effective.

4 Conclusions

PraK V4 advances interactive video search with spatially conjunctive localized queries, AVS-oriented re-ranking, richer intra-video exploration, a parallelized backend, and a hybrid keyframe layout, addressing bottlenecks from prior editions. Our annotation study shows that localized querying—especially via static regions or dynamic fusion—outperforms whole-image search, and that cross-region conjunction is crucial for difficult shots. In practice, faster within-video tools and low-latency infrastructure reduce misses and decision time in VBS 2026 tasks. Next, we will tighten region detection and the new online learning strategies with user studies during and after the competition.

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References

1. Amato, G., et al.: VISIONE 5.0: enhanced user interface and ai models for VBS2024. In: MMM, pp. 332–339 (2024). https://doi.org/10.1007/978-3-031-53302-0_29
2. Amato, G., et al.: VISIONE 5.0: toward evaluation with novice users. In: CBMI, pp. 1–6 (2024). <https://doi.org/10.1109/CBMI62980.2024.10859203>
3. Barthel, K.U., Hezel, N., Jung, K., Schall, K.: Improved evaluation and generation of grid layouts using distance preservation quality and linear assignment sorting (2022). <https://arxiv.org/abs/2205.04255>
4. Del Fabro, M., Münzer, B., Böszörményi, L.: Smart video browsing with augmented navigation bars. In: Advances in Multimedia Modeling, pp. 88–98 (2013)
5. Gia, B.T., et al.: NII-UIT at VBS2025: multimodal video retrieval with LLM integration and dynamic temporal search. In: MMM, pp. 318–325 (2025)
6. Heller, S., et al.: Interactive video retrieval evaluation at a distance: comparing sixteen interactive video search systems in a remote setting at the 10th video browser showdown. IJMIR **11**(1), 1–18 (2022). <https://doi.org/10.1007/S13735-021-00225-2>
7. Jäckl, B., Kloda, V., Keim, D.A., Lokoč, J.: Dynamic sub-region search in homogeneous collections using clip (2025). <https://doi.org/10.48550/arXiv.2506.09506>. to appear in SISAP 2025
8. Jäckl, B., Kloda, V., Keim, D.A., Lokoč, J.: Experimental evaluation of static image sub-region-based search models using clip (2025). <https://doi.org/10.48550/arXiv.2506.06938>. to appear in SISAP 2025
9. Jäckl, B., Kruchina, J., Joos, L., Keim, D.A., Peška, L., Lokoč, J.: Evaluating keyframe layouts for visual known-item search in homogeneous collections (2025)
10. Li, H., Mourad, A., Zhuang, S., Koopman, B., Zuccon, G.: Pseudo relevance feedback with deep language models and dense retrievers: successes and pitfalls (2022). <https://arxiv.org/abs/2108.11044>
11. Lokoč, J., et al.: A W2VV++ case study with automated and interactive text-to-video retrieval. In: ACM MM, pp. 2553–2561 (2020). <https://doi.org/10.1145/3394171.3414002>
12. Lokoc, J., Kovalčík, G., Soucek, T., Moravec, J., Cech, P.: A framework for effective known-item search in video. In: ACM MM, pp. 1777–1785 (2019). <https://doi.org/10.1145/3343031.3351046>
13. Lokoc, J., Mejzlík, F., Veselý, P., Soucek, T.: Enhanced SOMHunter for known-item search in lifelog data. In: ICMR, pp. 71–73 (2021). <https://doi.org/10.1145/3463948.3469074>
14. Lokoc, J., Peska, L.: A study of a cross-modal interactive search tool using CLIP and temporal fusion. In: MMM, vol. 13833, pp. 397–408 (2023). https://doi.org/10.1007/978-3-031-27077-2_31
15. Lokoč, J., et al.: Interactive video retrieval in the age of effective joint embedding deep models: lessons from the 11th VBS. Multimedia Syst. **29**(6), 3481–3504 (2023). <https://doi.org/10.1007/s00530-023-01143-5>
16. Mironică, I., Vertan, C., Ionescu, B.: A relevance feedback approach to video genre retrieval. In: IEEE ICCP, pp. 327–330 (2011). <https://doi.org/10.1109/ICCP.2011.6047890>
17. Nasirihaghighi, S., et al.: GynSurg: a comprehensive gynecology laparoscopic surgery dataset (2025). <https://arxiv.org/abs/2506.11356>

18. Nixon, L.J., et al.: Video retrieval for multimedia verification of breaking news on social networks. In: Proceedings of the First International Workshop on Multimedia Verification, pp. 13–21 (2017). <https://doi.org/10.1145/3132384.3132386>
19. Radford, A., et al.: Learning transferable visual models from natural language supervision. In: ICML, vol. 139, pp. 8748–8763 (2021)
20. Ren, T., et al.: Grounded SAM: assembling open-world models for diverse visual tasks (2024). <https://doi.org/10.48550/arXiv.2401.14159>
21. Rocchio, J.J.: Relevance feedback in information retrieval. In: The Smart Retrieval System - Experiments in Automatic Document Processing, pp. 313–323 (1971)
22. Rossetto, L., Schuldt, H., Awad, G., Butt, A.A.: V3C - a research video collection. In: Kompatsiaris, I., Huet, B., Mezaris, V., Gurrin, C., Cheng, W.H., Vrochidis, S. (eds.) MMM, pp. 349–360. Springer International Publishing, Cham (2019)
23. Schall, K., Barthel, K.U., Hezel, N., Jung, K.: Optimizing CLIP models for image retrieval with maintained joint-embedding alignment. In: SISAP, vol. 15268, pp. 97–110 (2024). https://doi.org/10.1007/978-3-031-75823-2_9
24. Schall, K., Hezel, N., Jung, K., Barthel, K.U.: Vibro: video browsing with semantic and visual image embeddings. In: MMM, pp. 665–670. Cham (2023)
25. Schoeffmann, K., Boeszoermyenyi, L.: Enhancing seeker-bars of video players with dominant color rivers. In: Advances in Multimedia Modeling, pp. 773–775 (2010)
26. Soucek, T., Lokoc, J.: TransNet V2: an effective deep network architecture for fast shot transition detection, MM '24, pp. 11218–11221. Association for Computing Machinery, New York (2024). <https://doi.org/10.1145/3664647.3685517>
27. Stroh, M., et al.: PraK tool V3: enhancing video item search using localized text and texture queries. In: MMM, pp. 326–333. Springer (2025)
28. Truong, Q.T., et al.: Marine video kit: a new marine video dataset for content-based analysis and retrieval. In: MMM, vol. 13833, pp. 539–550. Springer (2023). https://doi.org/10.1007/978-3-031-27077-2_42
29. Vadicamo, L., et al.: Evaluating performance and trends in interactive video retrieval: insights from the 12th VBS competition. IEEE Access **12**, 79342–79366 (2024). <https://doi.org/10.1109/ACCESS.2024.3405638>
30. Wang, X.: Intelligent multi-camera video surveillance: a review. Pattern Recogn. Lett. **34**(1), 3–19 (2013). <https://doi.org/10.1016/j.patrec.2012.07.005>